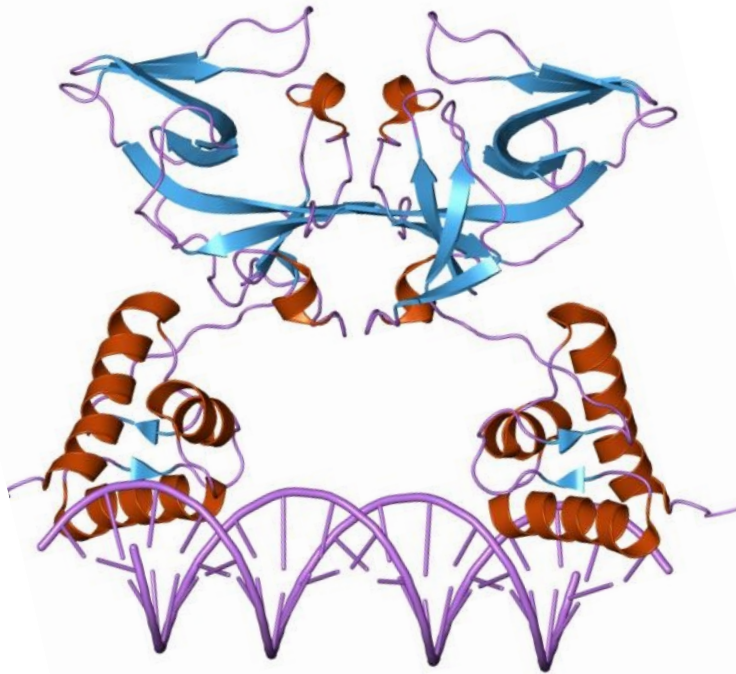


***Discovering phylogenetic footprints
in bacterial promoters***



- Example of DNA-protein interface.
 - LexA homodimer on the *recA* promoter.
 - Note: this is only a model, the structure has still not been crystallized.
 - Source: <http://srs.ebi.ac.uk>

LexA binding sites (RegulonDB, oct 2006)

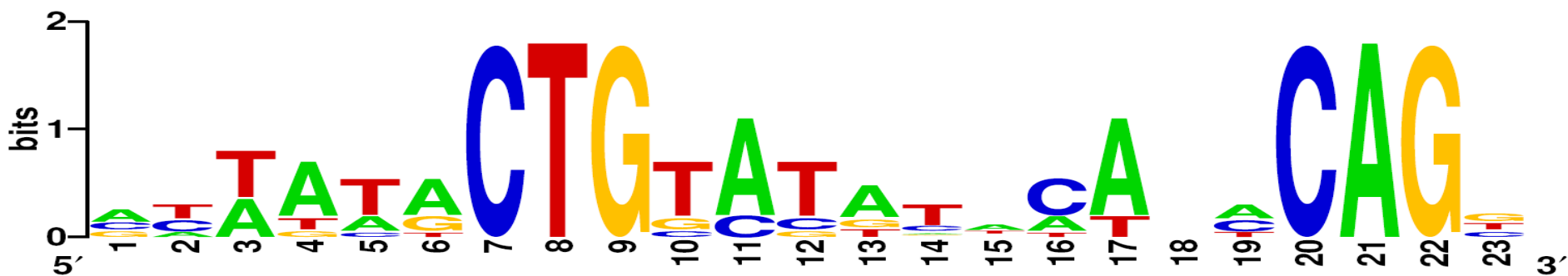
ECK120012770	recA	b2699	-21	gaagcaattaTAC CTGT ATGCTCATAC CAG Tatcaagtgttt
ECK120012770	lexA	b4043	-10	aatcgccttTTG CTGT ATATACTCAC CAG Cataactgtat
ECK120012770	lexA	b4043	10	atactcacagCATAA CTGT ATATACACCC CAG gggggcggaa
ECK120012770	recA	b2699	-21	gaagcaattaTAC CTGT ATGCTCATAC CAG Tatcaagtgttt
ECK120012770	ssb	b4059	-46.5	gacacaaattGAC CTGA ATGAATATAC CAG Tattggaatgc
ECK120012770	sulA	b0958	-2	agcccctgtgAGTTA CTGT ATGGATGTAC CAG tacatccag
ECK120012770	uvrA	b4058	-31.5	gacacaaattGAC CTGA ATGAATATAC CAG Tattggaatgc
ECK120012770	uvrB	b0779	-21	ttatggtgatGAA CTGT TTTTTTTATC CAG Tataatttgtt
ECK120012770	uvrD	b3813	11	taatcagcaaAT CTGT ATATATACCC CAG CTtttttggcgga
ECK120012770	rpsU	b3065	4.5	atthttgaaatAAG CTGG CGTTGATGCC CAG Cggcaaaccga
ECK120012770	phr	b0708	25.5	ttatcctgacGCC CTGG CTTTCAGGG CAG CGttatttcgaa

LexA aligned binding sites and position-specific scoring matrix (PSSM)

1 1	:	1/7	ATTATAC CT GTATGCTCATA CAGT
2 2	:	2/8	CTTTT GCT GTATATACTCA CAGC
3 3	:	3/10	GCATAA CT GTATATACACC CAGG
4 4	:	4/7	ATTATAC CT GTATGCTCATA CAGT
5 5	:	-5/13	CCAATA CT GTATATTCATT CAGG
6 6	:	-6/15	GATGTAC CT GTACATCCATA CAGT
7 7	:	-7/13	CCAATA CT GTATATTCATT CAGG
8 8	:	-8/13	ATTATAC CT GGATAAAAAA CAGT
9 9	:	9/7	GCAAAT CT GTATATATAC CAGC
10 10	:	10/8	AATAAG CT GGCGTTGATGCC CAGC
11 11	:	-11/12	ATAACG CT GCCCTGAAAGC CAGG



Pos	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
A	5	2	5	8	3	7	0	0	0	0	9	0	7	1	5	3	9	1	5	0	11	0	0
C	3	4	0	0	1	0	11	0	0	1	2	2	0	2	1	7	0	3	4	11	0	0	3
G	3	0	0	1	0	3	0	0	11	2	0	1	2	1	1	0	0	2	0	0	0	11	4
T	0	5	6	2	7	1	0	11	0	8	0	8	2	7	4	1	2	5	2	0	0	0	4



weblogo.berkeley.edu

The site sequences were obtained from RegulonDB.
The alignment and matrix were created using consensus (Hertz & Stormo, 1999).
The logo was created with Web Logo (<http://weblogo.berkeley.edu/logo.cgi>).

***Phylogenetic footprint discovery
in promoters of orthologous genes***

■ Orthologs of *Escherichia coli* K12 gene *PUTA* in Gammaproteobacteria

NP_245526.1	<i>Pasteurella multocida</i>	NP_415534.1	52.93	0.0
YP_340762.1	<i>Pseudoalteromonas haloplanktis</i> _TAC125	NP_415534.1	45.21	0.0
YP_271059.1	<i>Colwellia psychrerythraea</i> _34H	NP_415534.1	43.81	0.0
NP_719311.1	<i>Shewanella oneidensis</i>	NP_415534.1	47.35	0.0
YP_156342.1	<i>Idiomarina loihiensis</i> _L2TR	NP_415534.1	45.44	0.0
YP_095723.1	<i>Legionella pneumophila</i> _Philadelphia_1	NP_415534.1	51.25	0.0
YP_126994.1	<i>Legionella pneumophila</i> _Lens	NP_415534.1	51.63	0.0
YP_123979.1	<i>Legionella pneumophila</i> _Paris	NP_415534.1	51.25	0.0
NP_819659.1	<i>Coxiella burnetii</i>	NP_415534.1	47.21	0.0
YP_133644.1	<i>Photobacterium profundum</i> _SS9	NP_415534.1	46.63	0.0
YP_206789.1	<i>Vibrio fischeri</i> _ES114	NP_415534.1	48.96	0.0
NP_937700.1	<i>Vibrio vulnificus</i> _YJ016	NP_415534.1	47.40	0.0
NP_763030.1	<i>Vibrio vulnificus</i> _CMCP6	NP_415534.1	47.60	0.0
NP_801236.1	<i>Vibrio parahaemolyticus</i>	NP_415534.1	47.92	0.0
NP_639180.1	<i>Xanthomonas campestris</i>	NP_415534.1	52.40	0.0
YP_202784.1	<i>Xanthomonas oryzae</i> _KACC10331	NP_415534.1	52.69	0.0
YP_244967.1	<i>Xanthomonas campestris</i> _8004	NP_415534.1	52.40	0.0
NP_644196.1	<i>Xanthomonas citri</i>	NP_415534.1	52.59	0.0
YP_365739.1	<i>Xanthomonas campestris</i> _vesicatoria	NP_415534.1	52.40	0.0
NP_805570.1	<i>Salmonella typhi</i> _Ty2	NP_415534.1	91.29	0.0
NP_455618.1	<i>Salmonella typhi</i>	NP_415534.1	91.29	0.0
YP_052304.1	<i>Erwinia carotovora</i> _atrosepticaa	NP_415534.1	74.24	0.0
YP_408453.1	<i>Shigella boydii</i> _Sb227	NP_415534.1	99.09	0.0
YP_309997.1	<i>Shigella sonnei</i> _Ss046	NP_415534.1	99.39	0.0
NP_753076.1	<i>Escherichia coli</i> _CFT073	NP_415534.1	99.39	0.0
NP_415534.1	<i>Escherichia coli</i> _K12	NP_415534.1	100.00	0.0
NP_309287.1	<i>Escherichia coli</i> _O157H7	NP_415534.1	99.55	0.0
NP_287019.1	<i>Escherichia coli</i> _O157H7_EDL933	NP_415534.1	99.55	0.0
NP_871437.1	<i>Wigglesworthia brevipalpis</i>	NP_415534.1	54.57	0.0
NP_992900.1	<i>Yersinia pestis</i> _biovar_Mediaevails	NP_415534.1	79.44	0.0
NP_669761.1	<i>Yersinia pestis</i> _KIM	NP_415534.1	79.37	0.0
YP_070249.1	<i>Yersinia pseudotuberculosis</i> _IP32953	NP_415534.1	79.37	0.0
NP_405415.1	<i>Yersinia pestis</i> _CO92	NP_415534.1	79.37	0.0
NP_929224.1	<i>Photorhabdus luminescens</i>	NP_415534.1	77.37	0.0
YP_343707.1	<i>Nitrosococcus oceani</i> _ATCC_19707	NP_415534.1	50.00	0.0
YP_170117.1	<i>Francisella tularensis</i> _tularensis	NP_415534.1	50.34	0.0
YP_433088.1	<i>Hahella chejuensis</i> _KCTC_2396	NP_415534.1	48.61	0.0
NP_794750.1	<i>Pseudomonas syringae</i>	NP_415534.1	73.64	0.0
NP_747050.1	<i>Pseudomonas putida</i> _KT2440	NP_415534.1	73.45	0.0
YP_257639.1	<i>Pseudomonas fluorescens</i> _Pf-5	NP_415534.1	73.60	0.0
YP_346185.1	<i>Pseudomonas fluorescens</i> _PfO-1	NP_415534.1	73.45	0.0
NP_249473.1	<i>Pseudomonas aeruginosa</i>	NP_415534.1	48.56	0.0
YP_272798.1	<i>Pseudomonas syringae</i> _phaseolicola	NP_415534.1	73.77	0.0
YP_233614.1	<i>Pseudomonas syringae</i> _pv_B728a	NP_415534.1	73.80	0.0
YP_264533.1	<i>Psychrobacter arcticum</i> _273-4	NP_415534.1	44.42	0.0
YP_046314.1	<i>Acinetobacter</i> _sp_ADPI	NP_415534.1	59.51	0.0

Upstream sequences of PUTA orthologs

>YP_046314.1|Acinetobacter_sp_ADPl|putA YP_046314.1; upstream from -103 to -1; size: 103;
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ATGATAAAATTATCTAAACCCCTTTAAACAATATACCTTAGAGT
>YP_271059.1|Colwellia_psychrerythraea_34H|putA YP_271059.1; upstream from -245 to -1; size: 245
ATAATAACCCACGAACTCCCTACAAATTATAAAAAACGATTTGCAGCACTTTTATACTG
TTGAATTCGTACTCCCCCATATAAAAGTGTTTTAAATCCTGAAAAATAACCAGCACATCCT
GTGGTTGTTACCCTAAAAATTCGCTCATAAAATTAATGTCGTCACCAACTAATAATATATG
TATTAGTGGAAAAAAGACTATAACTAAAAGCAGGATTTCTACCTGTCACACTTTGAGGAA
TGGTT
>NP_819659.1|Coxiella_burnetii|putA NP_819659.1; upstream from -118 to -1; size: 118;
GAAGTAGCCCGTATGAAGCGAAGCGAAATACGGGGAGGTGCACGTATTGTTCCCGTATTC
GCTTCGTTTCATACGGGCTACATCGCGGAAATGAAAAATTAACCTCCTTAATGAGGACAT
>YP_052304.1|Erwinia_carotovora_atroseptica_SCR1043|putA YP_052304.1; upstream from -195 to -1; size: 195;
TTAACTCTCCACATTTTTTTTCGTGCCGCCGTCGCGGCGACGCTGTGTTTTATAGTAATCA
TTCAGGCCGGAACGAGGTCTGAAAAATGATTATGGGCAGCAACCATTCCATTTGTTAACAA
GGTTGCACAAAAGTTGCAACATGATTGATATTGACGGTATCCCGATGTGCATCTTCTCATT
ACAGGAGTGGACTCT
>NP_753076.1|Escherichia_coli_CFT073|putA NP_753076.1; upstream from 0 to -1; size: 0;
>NP_415534.1|Escherichia_coli_K12|putA NP_415534.1; upstream from -400 to -1; size: 400;
ATCGGCAATGTCGAAACTTGCCGTTATATCTGCCACCGGAACGGGGTAACAGAGTTTATG
TTTTACCAGGGCGACCGTATCCTGCCGGAAGCGCTGGTTATTTCACAATCGATTTAACACA
CCATTTACATTAAATTTTAGTGCTCAGCGACACTATTTTTTCATCAGGTTGCACTCTCTCA
CATTTTTTTGCGGTTGCACCTTTTCAAAAAATGTTAACTGCCGCAGAGAAAAAGTCTGAGTTA
TTTTTTTAAATCCCTGTCATATCGATTTCTTTTATTAACATTTTCATTCATTTTTTAAGCTTG
CTACGCATGTCACATTTAACATGGTTGCACAAAAGTTGCAACATCATGGATATTTACAGAT
AACGTTAAGTTGCACCTTTCTGAACAACAGGAGTAATGGC
>NP_309287.1|Escherichia_coli_O157H7|ECs1260 NP_309287.1; upstream from -54 to -1; size: 54;
GGATATTTTCACGATAACGTTAAGTTGCACCTTTTCAGAAACAACAGGAGTAATGGC
>NP_287019.1|Escherichia_coli_O157H7_EDL933|putA NP_287019.1; upstream from -400 to -1; size: 400;
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TTTTACCAGGGCGACCGTATCCTGCCGGAAGCGCTGGTTATTTCACAATCGATTTAACACA
CCATTTACATTAAATTTTAGTGCTCAGCGACACTATTTTTTCATCAGGTTGCACTCTCTCA
CATTTTTTTGCGGTTGCACCTTTTCAAAAAATGTTAACTGCCGCAGAGAAAAAGTCTGAGTTA
TTTTTTTAAATCCCTGTCATATCGATTTCTTTTATTAACATTTTCATTCATTTTTTAAGCTTG
CTACGCAGGTACATTTAACATGGTTGCACAAAAGTTGCAACATCATGGATATTTACAGAT
AACGTTAAGTTGCACCTTTTCAGAAACAACAGGAGTAATGGC

....

Purged upstream sequences of PUTA orthologs

[illegible]

...

Significantly over-represented dyads in promoters of PUTA orthologs

; column headers

1 sequence

2 identifier

3 expected_freq

4 occ observed occurrences

5 exp_occ expected occurrences

6 occ_P occurrence probability (binomial)

7 occ_E E-value for occurrences (binomial)

8 occ_sig occurrence significance (binomial)

9 ovl_occ number of overlapping occurrences

10 all_occ number of non-overlapping + overlapping occurrences

11 rank rank

12 ov_coef overlap coefficient

13 remark remark

sequence	identifier	expected_freq	occ	exp_occ	occ_P	occ_E	occ_sig	ovl_occ	all_occ	rank	ov_coef	remark
gtgn{1}aac	gtgn{1}aac gttn{1}cac	0.00058697	32	4.44	2.80E-14	1.10E-09	8.95	0	32	1	1.0166	
ggtn{3}acc	ggtn{3}acc ggtn{3}acc	0.00016832	18	1.26	2.60E-13	1.00E-08	7.99	0	18	2	1.0166	inv_rep
gttn{0}gca	gttn{0}gca tgc{0}aac	0.00077131	35	5.87	3.30E-13	1.30E-08	7.88	5	40	3	1.0166	
ggtn{2}aac	ggtn{2}aac gttn{2}acc	0.00048995	28	3.68	3.60E-13	1.40E-08	7.84	0	28	4	1.0166	
gtgn{0}caa	gtgn{0}caa ttgn{0}cac	0.00070077	32	5.34	2.80E-12	1.10E-07	6.95	0	32	5	1.0166	
ggtn{2}cac	ggtn{2}cac gtgn{2}acc	0.00040329	24	3.03	7.40E-12	2.90E-07	6.53	0	24	6	1.0166	
ggtn{0}gca	ggtn{0}gca tgc{0}acc	0.00052995	27	4.04	1.20E-11	4.90E-07	6.31	0	27	7	1.0166	
gcan{0}acc	gcan{0}acc ggtn{0}tgc	0.00052995	27	4.04	1.20E-11	4.90E-07	6.31	0	27	8	1.0166	
ggtn{1}caa	ggtn{1}caa ttgn{1}acc	0.00058494	27	4.42	1.10E-10	4.20E-06	5.38	0	27	9	1.0166	
ggtn{1}gca	ggtn{1}gca tgc{1}acc	0.00052995	24	4.01	1.60E-09	6.40E-05	4.20	0	24	10	1.0166	
aagn{2}gca	aagn{2}gca tgc{2}ctt	0.00062177	26	4.67	1.70E-09	7.00E-05	4.16	1	27	11	1.0166	
aggn{1}gca	aggn{1}gca tgc{1}cct	0.0004801	22	3.63	6.10E-09	2.40E-04	3.62	0	22	12	1.0166	
aggn{0}tgc	aggn{0}tgc gcan{0}cct	0.0004801	20	3.66	1.30E-07	5.20E-03	2.28	0	20	13	1.0166	
aggn{2}caa	aggn{2}caa ttgn{2}cct	0.00052992	21	3.98	1.40E-07	5.80E-03	2.24	0	21	14	1.0166	
aagn{1}tgc	aagn{1}tgc gcan{1}ctt	0.00062177	20	4.7	6.30E-06	2.50E-01	0.60	1	21	15	1.0166	
aggn{3}aac	aggn{3}aac gttn{3}cct	0.00044387	16	3.32	1.30E-05	5.30E-01	0.28	0	16	16	1.0166	
gggn{4}acc	gggn{4}acc ggtn{4}ccc	0.00025997	12	1.93	1.50E-05	5.90E-01	0.23	4	16	17	1.0166	

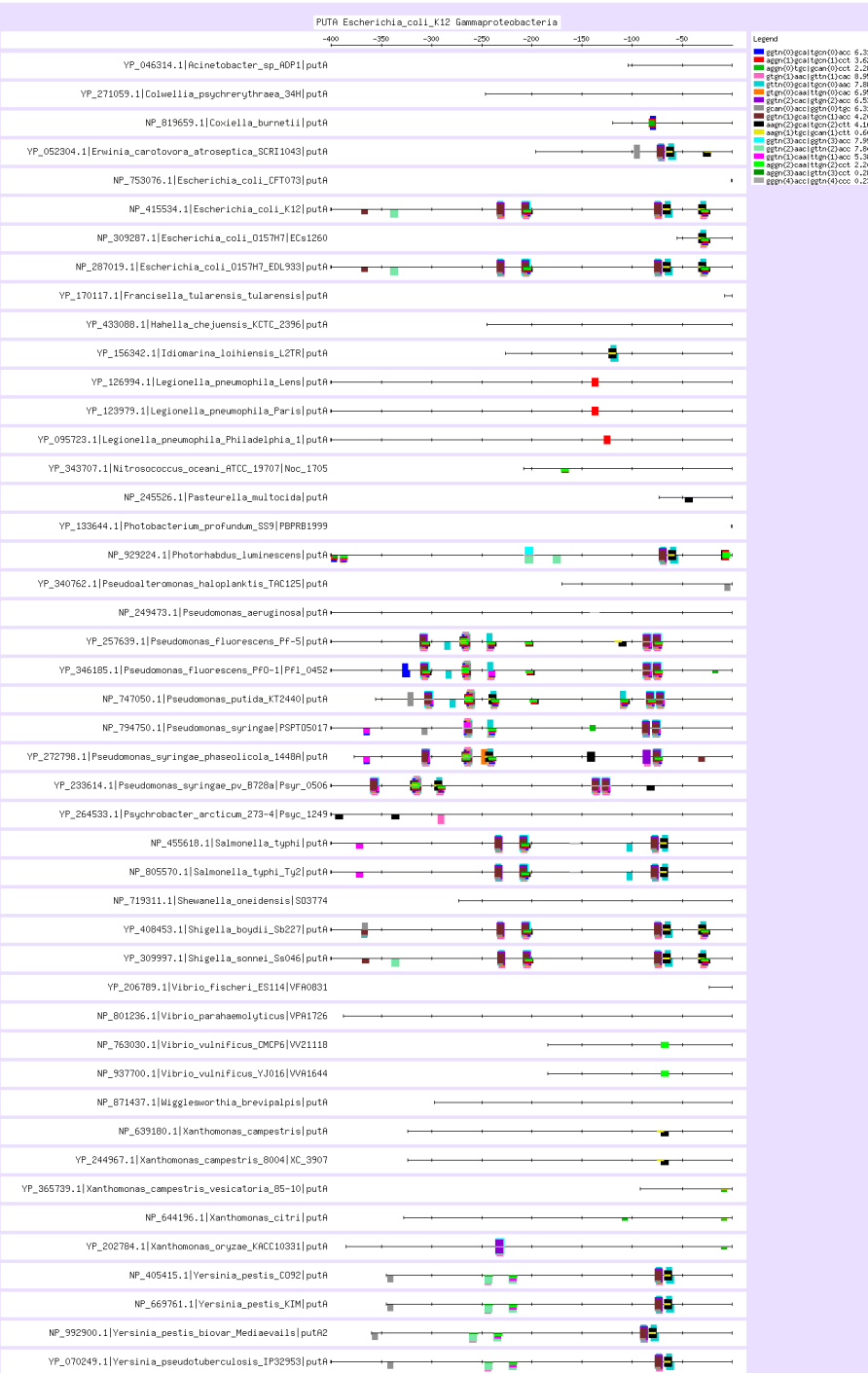
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;Job done 13/02/06 21:31:28 CET

Significantly over-represented dyads in promoters of PUTA orthologs

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; pattern-assembly -v 1 -i /home/jvanheld/research/collabo
; Input score column      8
; Output score column    0
; two strand assembly
; max flanking bases     1
; max substitutions      0
; max assembly size      50
; max number of pattern  50
; number of input patter 17
;
```

assembly # 1	seed: gtgnaa	17 words	length
align	rev_cpl	score	
aagngca....	...tgcnnctt	4.16	
aagtngc.....	gcanctt	0.6	
.aggngca....	...tgcncct.	3.62	
.agggtgc.....	gcacct.	2.28	
.aggnncaa...	...ttgnnctt.	2.24	
.aggnnnaac..	..gttnnnctt.	0.28	
..ggtnnnacc.	.ggtnnnacc..	7.99	
..ggttnnaac..	..gttnnacc..	7.84	
..ggtgca....	...tgcacc..	6.31	
..ggttncaa...	...ttgnacc..	5.38	
..ggtnnnnccc	gggnnnnacc..	0.23	
...gtgnaac..	..gttnnac...	8.95	
...gtgcaa...	...ttgcac...	6.95	
...gtgnnacc.	.ggttnnac...	6.53	
....tgcaac..	..gttgca....	7.88	
....tgcnaac.	.ggtngca....	4.2	
....gcaacc.	.ggttgca....	6.31	
aagggtgcaacc	gggttgcacctt	8.95	best
Job started 13/02/06 21:31:29 CET			
Job done 13/02/06 21:31:31 CET			



- For each gene, we applied the same pattern discovery approach
 - Identify orthologs
 - Retrieve upstream sequences
 - Detect over-represented dyads
- Questions
 - How good is this method in predicting cis-acting elements ?
 - Can we detect correct motifs ?
 - What is the rate of false positives ?
 - Can we learn something about the evolution of cis-acting elements ?
 - On the basis of the discovered motifs, can we regroup the co-regulated genes ?
 - Detect pair-wise associations between genes
 - Detect clusters of genes regulated by the same transcription factor

***Evolution of the auto-regulation
of the LexA transcription factor***

Study case: LexA auto-regulation

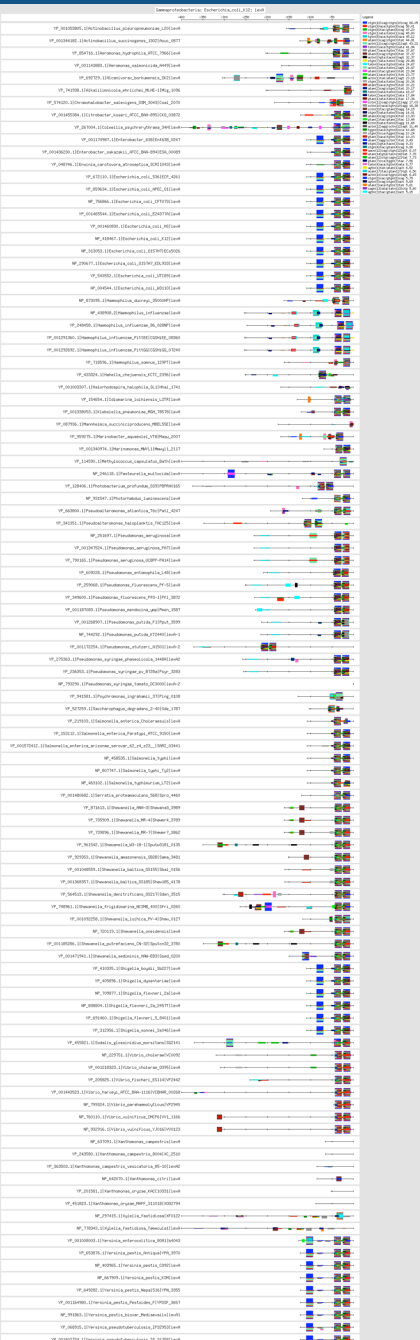
- The transcription factor LexA represses several genes involved in the SOS response
- The *lexA* gene is auto-regulated.
- LexA auto-regulation has been characterized in details in several bacterial species.
 - LexA protein has evolved and recognizes different motifs in different taxa.
- Note that this is an easy case:
 - LexA binding motif is highly conserved in Gammaproteobacteria
 - This motif is easily detected by most pattern discovery algorithms.
- We will start by this example, and then generalize the evaluation to other transcription factors.

Evolution of lexA binding motifs
Motifs characterized experimentally in different bacterial taxa

Taxonomic level	Species	Consensus	Weblogo
Gamma-proteobacteria	<i>E. coli</i>	CTGTNNNNNNNNACAG	
Xanthomonadales	<i>X. oryzae</i> <i>X. campestris</i> <i>X. citri</i> <i>Xylella fastidiosa</i>	TTAGTArwawTACTAa	
Alpha proteobacteria	<i>R. etli</i> <i>R. meliloti</i> <i>A. tumefaciens</i> <i>Rhodobacter sphaeroides</i> <i>Sinorhizobium meliloti</i>	G TTCNNNNNNNGTTC GAACNNNNNNNGAAC	
Delta proteobacteria	<i>Myxococcus xanthus</i>	CTRHAMRYBYGTT CAGS	
	<i>Geobacter sulfurreducens</i>	GGTTNNCNNNNNGNNNACC	
Cyanobacteria	<i>Anabaena PCC7120</i>	TAGTACTWATGTTCTA	
Gram+ bacteria	<i>B. subtilis</i> <i>Mycobacterium tuberculosis</i> <i>Mycobacterium smegmatis</i>	CGAACRNRYGTTYC	

Janky, R. and van Helden, J. (2008). Evaluation of phylogenetic footprint discovery for predicting bacterial cis-regulatory elements and revealing their evolution. *BMC Bioinformatics* 9, 37.

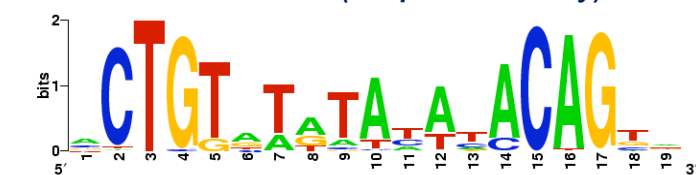
Significant dyads in promoters of *lexA* orthologs in Gammaproteobacteria



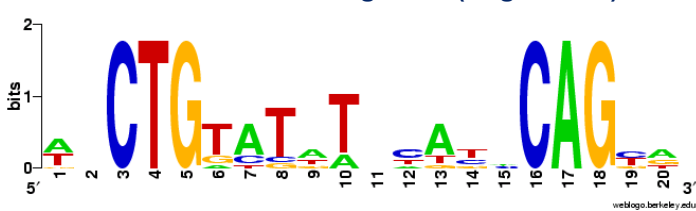
- In Gammaproteobacteria, the most significant motif is
aaCTGtatacatataCAGtt
- This corresponds to the LexA motif annotated in RegulonDB

Footprint-discovery result with
lexA in Gammaproteobacteria,
March 2009

Discovered motif (footprint-discovery)

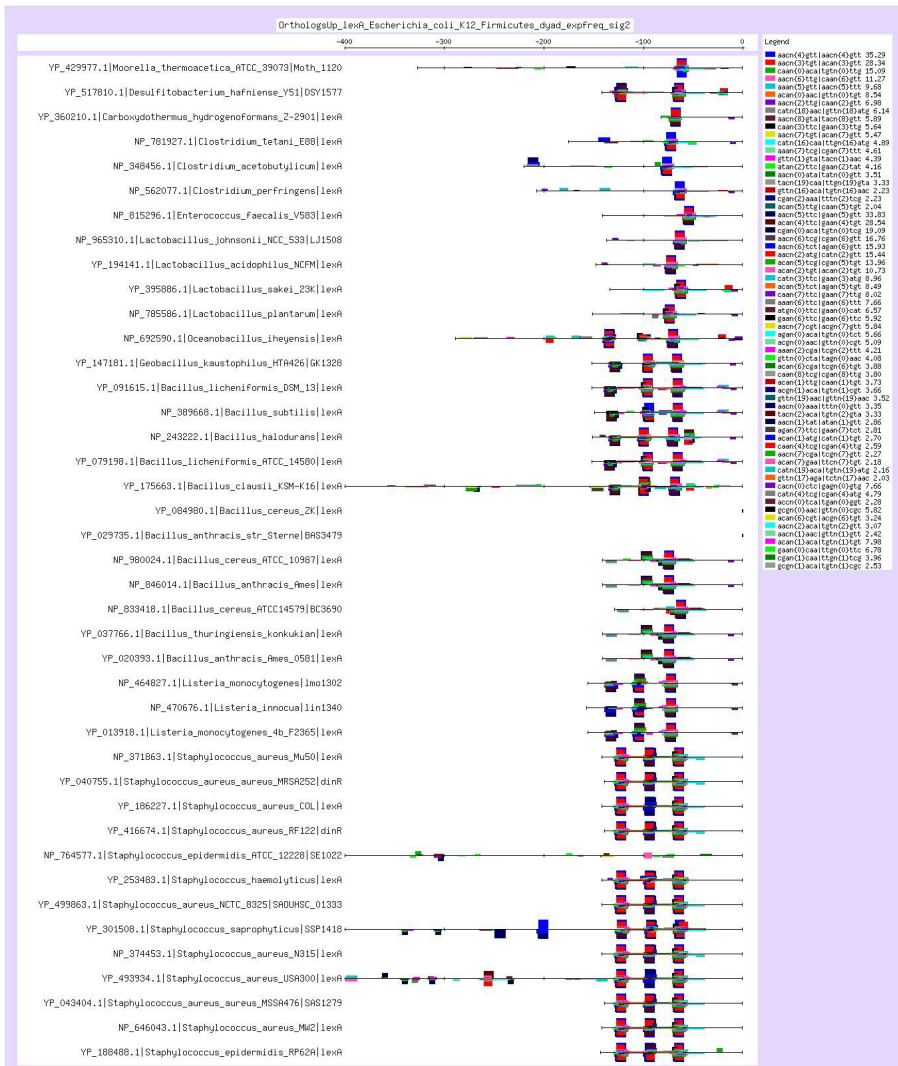


Annotated LexA binding motif (RegulonDB)



■ Janky,R. and van Helden,J. (2008) Evaluation of phylogenetic footprint discovery for predicting bacterial cis-regulatory elements and revealing their evolution. *BMC Bioinformatics*, 9, 37.

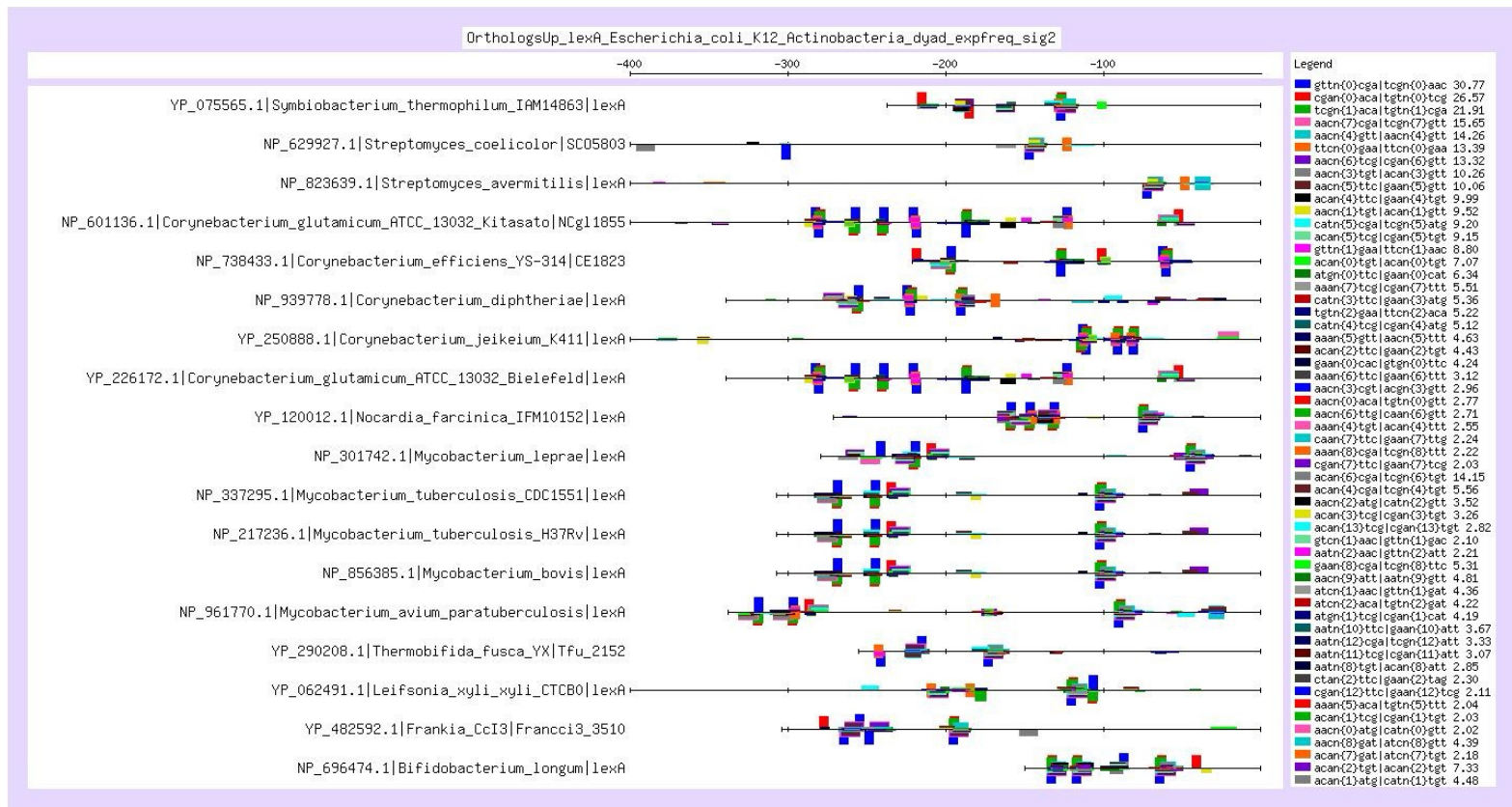
Significant dyads in promoters of *lexA* orthologs in Firmicutes



- In Firmicutes, the most significant motif is
TAC**GAAC**ATAT**GTTC**GTAA
- This corresponds to the “Cheo box”
GAAC_nGTTC

Significant dyads in promoters of *lexA* orthologs in Actinobacteria

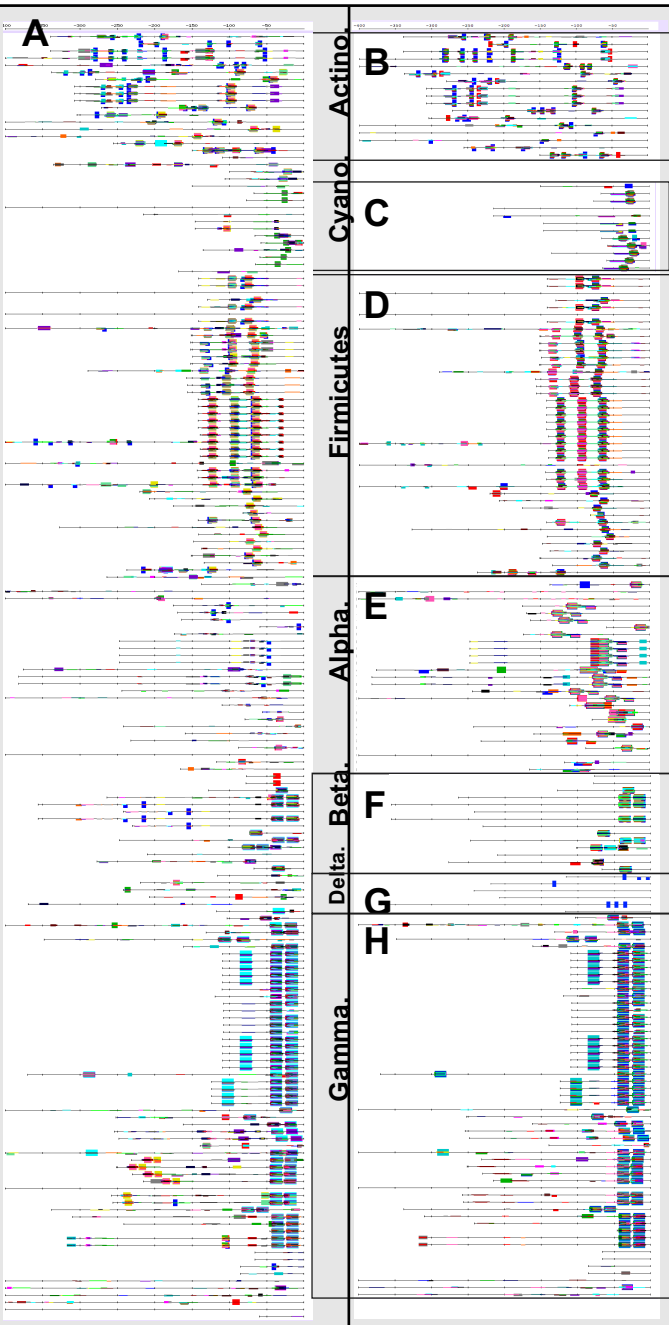
- In Actinobacteria, the most significant motif is **TCGAACA**.
- This shows an almost perfect match to one half of the Cheo box motif detected in Firmicutes (**TACGAACATATGTTTCGTA**)
- In addition, this larger Cheo box is also detected in Actinobacteria, albeit with a lower significance than the half site.



Divergence of the LexA binding motif



Rekin's Janky

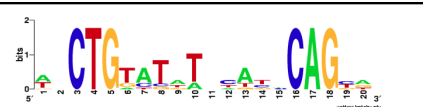
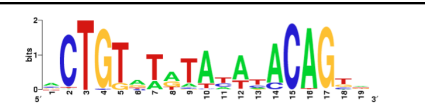
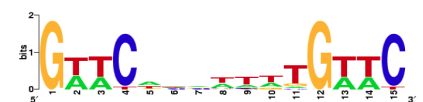
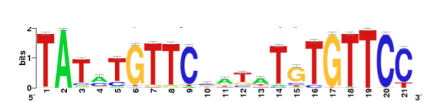
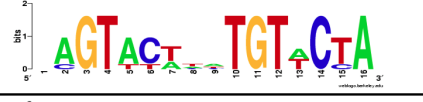
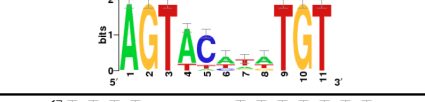
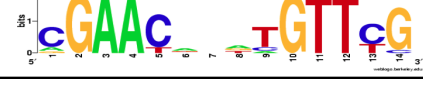
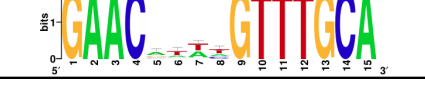


- Divergence of LexA binding motif
 - Footprint-discovery finds different motifs in LexA promoters of different taxa
- Robustness of the analysis
 - When all the bacterial promoters are analyzed together, the program dyad-analysis detects most of taxon-specific motifs discussed before, and the feature-map highlights their taxon-specific locations.
 - This illustrates the robustness of the method: the motifs can be detected even if present in a subset of the sequences only.
 - The significance is however lower when all sequences are analyzed together than with the taxon-per-taxon analysis.

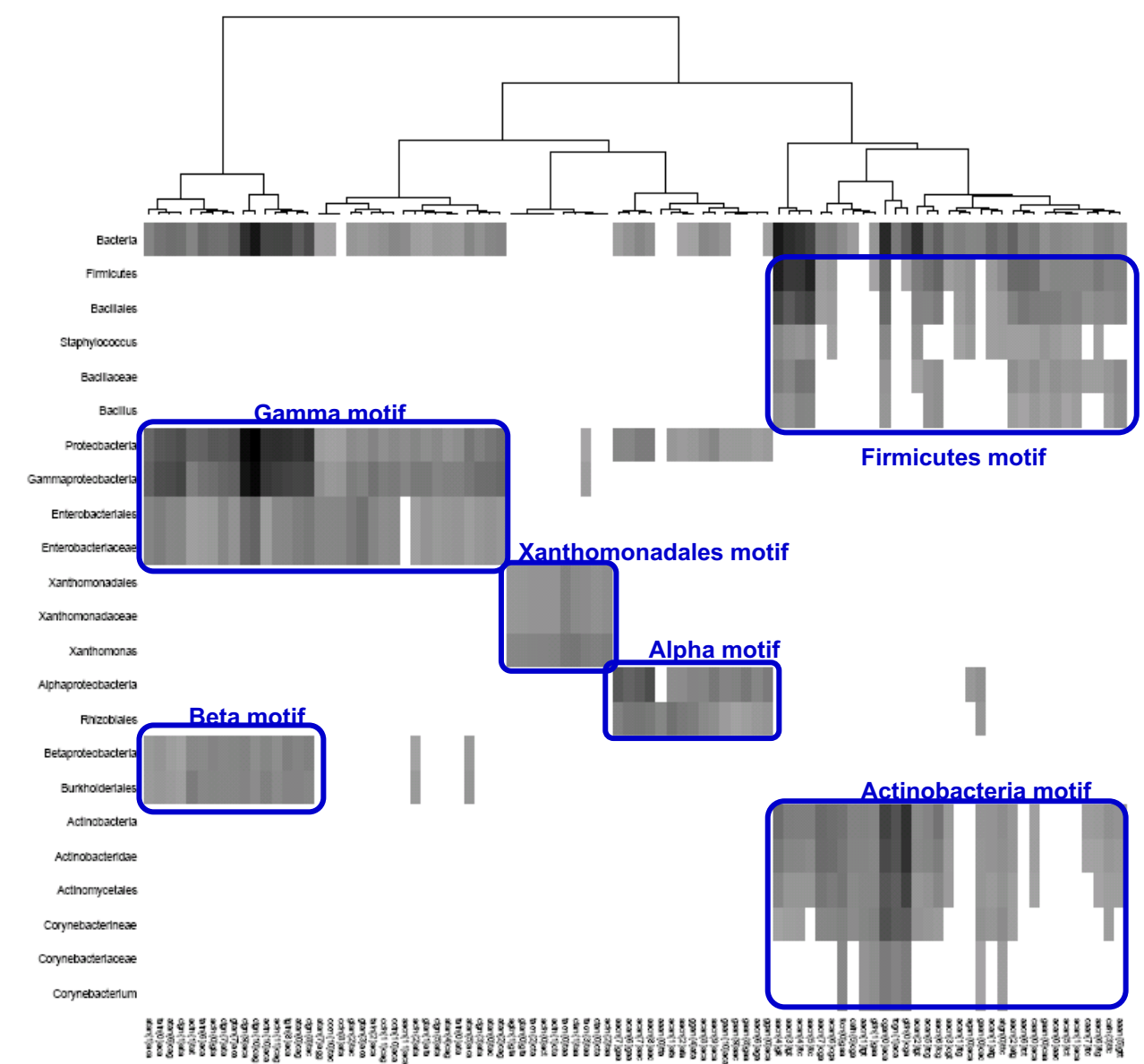
Gamma-proteobacteria	<i>E. coli</i>	CTGTNNNNNNNNACAG		
Xanthomonadales	<i>X. oryzae</i> <i>X. campestris</i> <i>X. citri</i> <i>Xylella fastidiosa</i>	TTAGTA ^{rwaw} TACTAa		gTTAGTAatacTACTAAc
Alpha proteobacteria	<i>R. etli</i> <i>R. meliloti</i> <i>A. tumefaciens</i> <i>Rhodobacter sphaeroides</i> <i>Sinorhizobium meliloti</i>	GTTNNNNNNNGTTC GAACNNNNNNNGAAC		
Delta proteobacteria	<i>Myxococcus xanthus</i>	CTRHAMRYBYGTTTCAGS ctRHAMRYbygttcags		GTAAGT
	<i>Geobacter sulfurreducens</i>	GGTTNNC ⁿⁿⁿⁿⁿⁿ NNNACC		
Cyanobacteria	<i>Anabaena PCC7120</i>	TAGTACTWATGTTCTA		
Gram+ bacteria	<i>B. subtilis</i> <i>Mycobacterium tuberculosis</i> <i>Mycobacterium smegmatis</i>	CGAACRNYGTTYC		

Footprint discovery retraces the evolution of LexA binding motif

- We ran footprint-discovery at different taxonomical levels.
- The discovered motifs vary depending on the taxon.
- All known LexA motifs are re-discovered.
- (table from Janky et al. 2008, motif discovery update 2012/01).

Gamma-proteobacteria	<i>E.coli</i>	CTGTNNNNNNNNACAG		
Xanthomonadales	<i>X.oryzae</i> <i>X.campestris</i> <i>X.citri</i> <i>Xylella fastidiosa</i>	TTAGTA ^r wawTACTAA		gTTAGTAatacTACTAAc
Alpha proteobacteria	<i>R.etli</i> <i>R.meliloti</i> <i>A.tumefasciens</i> <i>Rhodobacter sphaeroides</i> <i>Sinorhizobium meliloti</i>	G ^r TTCNNNNNNNGTTC GAACNNNNNNNGAAC		
Delta proteobacteria	<i>Myxococcus xanthus</i>	CTRHAMRYBYGTTTCAGS ctRHAMRYbygttcags		GTAAGT
	<i>Geobacter sulfurreducens</i>	GGTTNNCNNNGNNNACC		
Cyanobacteria	<i>Anabaena PCC7120</i>	TAGTACTWATGTTCTA		
Gram+ bacteria	<i>B.subtilis</i> <i>Mycobacterium tuberculosis</i> <i>Mycobacterium smegmatis</i>	CGAACRNRYGTTYC		

Significance map of the motifs discovered in promoters of *lexA* orthologs at all taxonomical levels

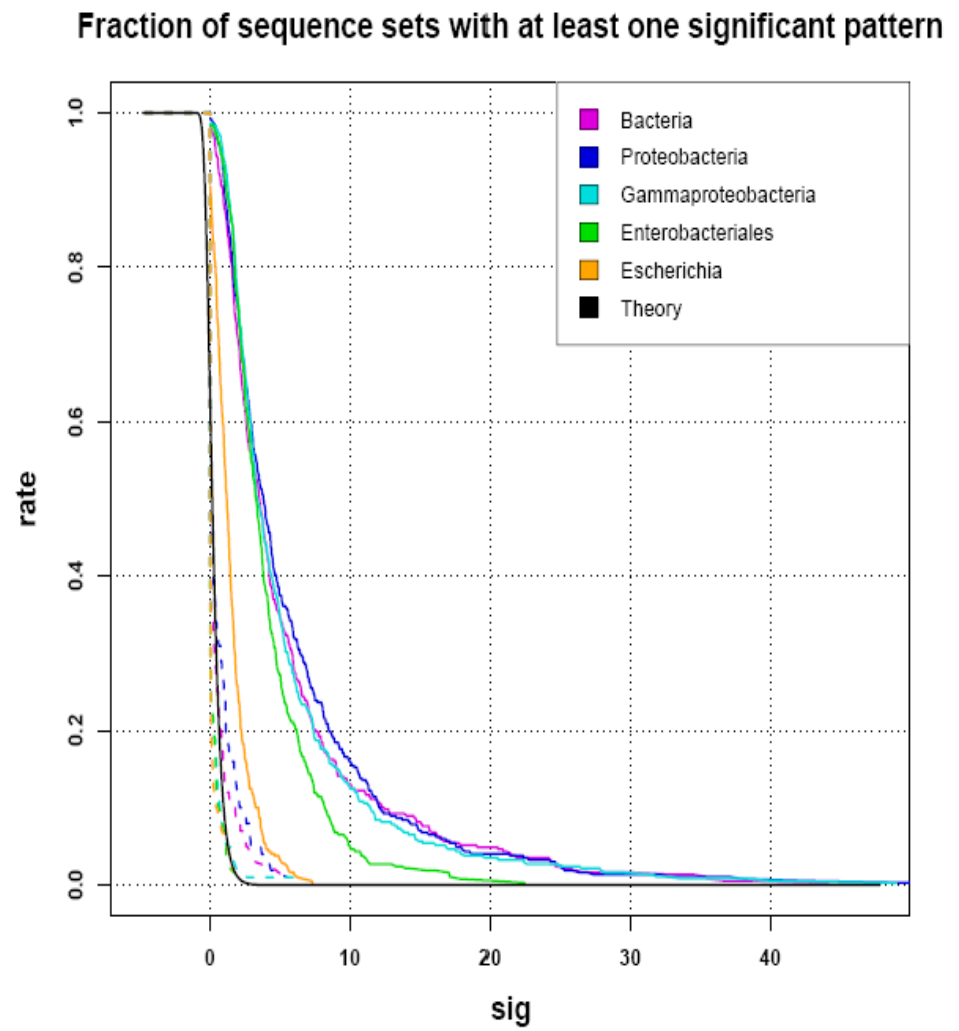
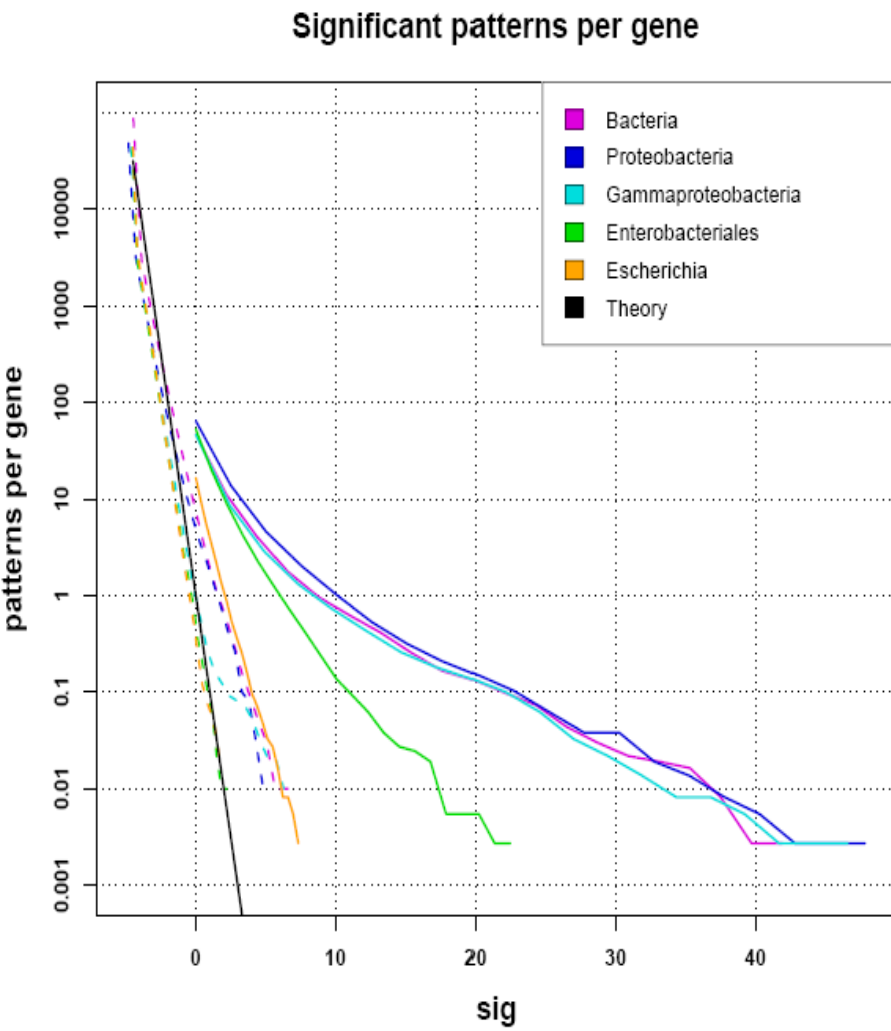


- The heat map illustrates the most significant motifs (sig ≥ 8) found at different taxonomical levels.
- Each column corresponds to one dyad, each row to one taxon.
- The grey level indicates the significance.

***Systematic evaluation of
phylogenetic footprints discovery***

- Choice of the sequence sets:
 - Positive set: promoters of orthologs for all the genes having at least one annotated site in RegulonDB
 - Negative set: random selections of promoters.
- Evaluation criteria
 - Distribution of significance scores: does this score allow us to discriminate groups of co-regulated promoters from random selections?
 - Motif correctness: do the motifs predicted in the positive set correspond to the annotated sites ?

Evaluation of significance scores: distribution in promoters of orthologs versus random selections of genes



- How to compare a discovered motif with a collection of annotated binding sites ?
 - The annotated binding sites can be considered as a set of oligonucleotides (“words”).
 - The discovered motif is a set of dyads.
 - We can compare two sets of words with the program compare-patterns.

Annotated sites for the gene glpD (Escherichia coli K12)

Site sequence	Factor	site ID
gataaacgccATAATGTTATACATATCACTCTaaaatgtttt	CRP	ECK120014013
tcttttgctaaTATGTTTCGATAACGAACATTtatgagcttt	GlpR	ECK120012732
taacgaacatTTATGAGCTTTAACGAAAGTgaatgagggc	GlpR	ECK120012734
gggatcaactGGTTTGCGTTTTGGCGCAAAttcagtgtta	GlpR	ECK120013968
aaaccggaaaTTAAGCGCGGATTTCGAATATtctgactggt	GlpR	ECK120013970

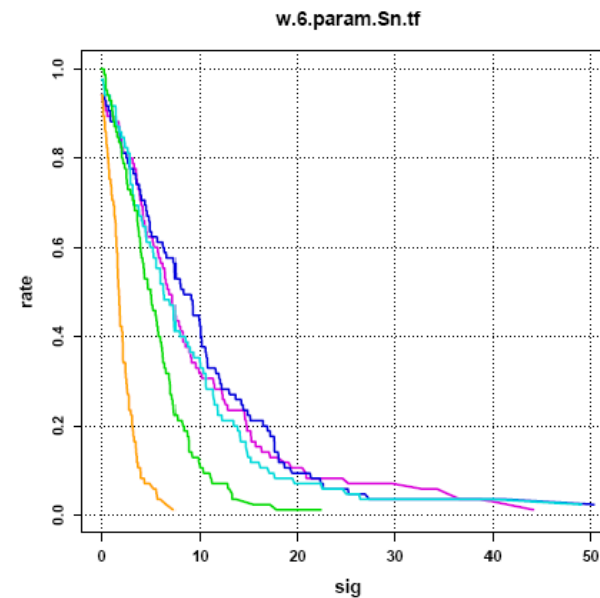
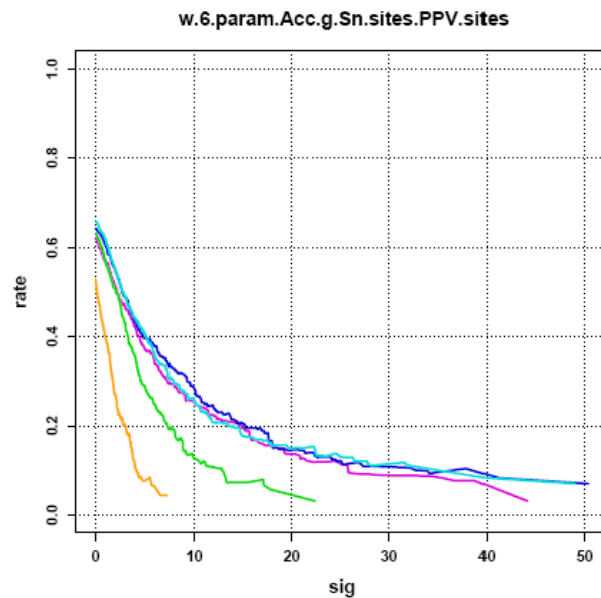
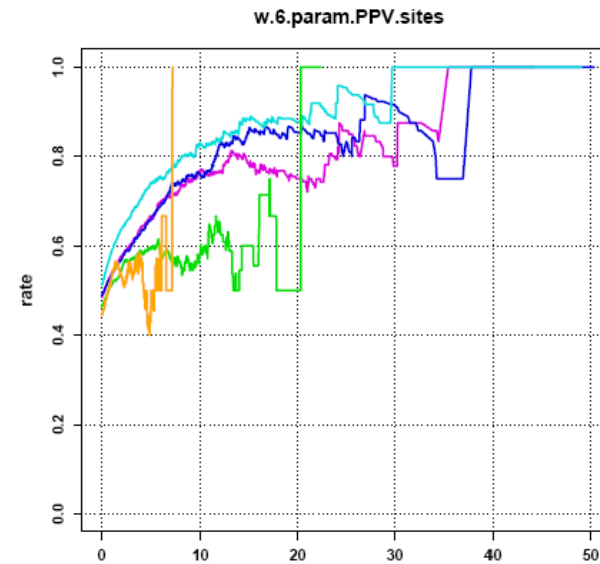
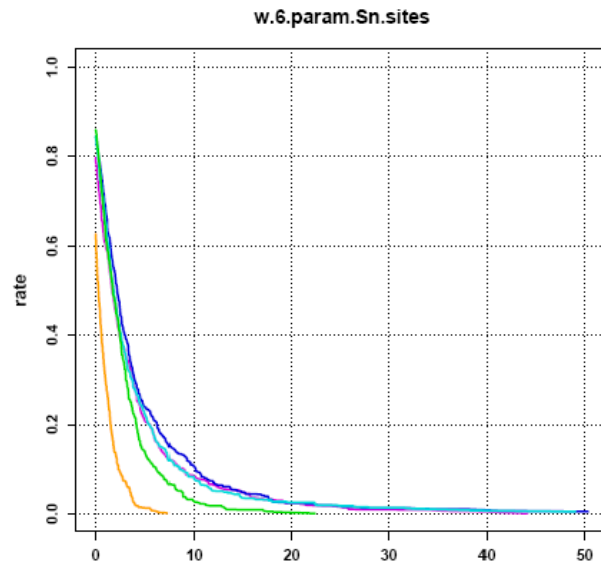
Perfect matches between detected dyads and annotated sites

Site ID	matching bases	strand	offset	weight	seq1	seq2
CRP_glpD_ECK120014013		6 R	30	6	AAAACATTTTAGAGTGATATGTATAACATTATGGCGTTTATC	aacn{0}att
GlpR_glpD_ECK120012732		7 D	11	6	tcttttgctaaTATGTTTCGATAACGAACATTtatgagcttt	atgn{1}tcg
GlpR_glpD_ECK120012732		6 D	24	6	tcttttgctaaTATGTTTCGATAACGAACATTtatgagcttt	aacn{0}att
GlpR_glpD_ECK120012732		6 D	22	6	tcttttgctaaTATGTTTCGATAACGAACATTtatgagcttt	cgan{0}aca
GlpR_glpD_ECK120012732		8 D	20	6	tcttttgctaaTATGTTTCGATAACGAACATTtatgagcttt	aacn{2}aca
GlpR_glpD_ECK120012732		8 R	20	6	AAAGCTCATAAATGTTTCGTTATCGAACATATTAGCAAAGA	aatn{2}tcg
GlpR_glpD_ECK120012732		6 D	11	6	tcttttgctaaTATGTTTCGATAACGAACATTtatgagcttt	atgn{0}ttc
GlpR_glpD_ECK120012732		8 D	21	6	tcttttgctaaTATGTTTCGATAACGAACATTtatgagcttt	acgn{2}cat
GlpR_glpD_ECK120012732		9 D	20	6	tcttttgctaaTATGTTTCGATAACGAACATTtatgagcttt	aacn{3}cat
GlpR_glpD_ECK120012732		6 D	20	6	tcttttgctaaTATGTTTCGATAACGAACATTtatgagcttt	aacn{0}gaa
GlpR_glpD_ECK120012734		7 R	0	6	GCCCTCATTCACTTTTCGTTAAAGCTCATAAATGTTTCGTTA	atgn{1}tcg
GlpR_glpD_ECK120012734		6 D	5	6	taacgaacatTTATGAGCTTTAACGAAAGTgaatgagggc	aacn{0}att
GlpR_glpD_ECK120012734		6 D	3	6	taacgaacatTTATGAGCTTTAACGAAAGTgaatgagggc	cgan{0}aca
GlpR_glpD_ECK120012734	19 D	6	6	6	taacgaacatTTATGAGCTTTAACGAAAGTgaatgagggc	acan{13}acg
GlpR_glpD_ECK120012734	8 D	1	6	6	taacgaacatTTATGAGCTTTAACGAAAGTgaatgagggc	aacn{2}aca
GlpR_glpD_ECK120012734	8 R	1	6	6	GCCCTCATTCACTTTTCGTTAAAGCTCATAAATGTTTCGTTA	aatn{2}tcg
GlpR_glpD_ECK120012734	6 R	0	6	6	GCCCTCATTCACTTTTCGTTAAAGCTCATAAATGTTTCGTTA	atgn{0}ttc
GlpR_glpD_ECK120012734	8 D	2	6	6	taacgaacatTTATGAGCTTTAACGAAAGTgaatgagggc	acgn{2}cat
GlpR_glpD_ECK120012734	9 D	1	6	6	taacgaacatTTATGAGCTTTAACGAAAGTgaatgagggc	aacn{3}cat
GlpR_glpD_ECK120012734	10 R	28	6	6	GCCCTCATTCACTTTTCGTTAAAGCTCATAAATGTTTCGTTA	cctn{4}cac
GlpR_glpD_ECK120012734	6 D	1	6	6	taacgaacatTTATGAGCTTTAACGAAAGTgaatgagggc	aacn{0}gaa
GlpR_glpD_ECK120012734	13 R	20	6	6	GCCCTCATTCACTTTTCGTTAAAGCTCATAAATGTTTCGTTA	actn{7}aag
GlpR_glpD_ECK120013970	8 R	21	6	6	AACAGTCAGAATATTCGAATCCGCGCTTAATTTCCGGTTT	aatn{2}tcg
GlpR_glpD_ECK120013970	14 D	1	6	6	aaaccggaaaTTAAGCGCGGATTTCGAATATtctgactggt	aacn{8}aag

- The dyad-site cross-table allows us to define correspondence statistics.
- Sensitivity (Sn)
 - $Sn = TP/(TP+FN)$
 - fraction of annotated sites matched by at least one discovered dyad
- Positive Predictive Value (PPV)
 - $PPV = TP/(TP+FP)$
 - Fraction of discovered dyads matching at least one annotated site
- Accuracy (geometric)
 - $Acc=sqrt(Sn \cdot PPV)$
 - Geometric mean of Sn and PPV

	gataaacgccATAATGTTATACATATCACTCTaaaaatgtttt	tctttgctaaTATGTTTCGATAACGAACATTtatgagcttt	taacgaacatTTATGAGCTTTAACGAAAGTgaatgagggc	gggatcaactGGTTTGCGTTTTGGCGCAAAAttcagtgta	aaaccggaaaTTAAGCGCGGATTTCGAATATtctgactgtt
; sequence					
atgn{1}tcg	.	o	o	.	.
aacn{0}att	o	o	o	.	.
cgan{0}aca	.	o	o	.	.
agtn{10}tcg	.	.	.	.	.
acan{13}acg	.	.	o	.	.
aacn{2}aca	.	o	o	.	.
aatn{2}tcg	.	o	o	.	o
atgn{0}ttc	.	o	o	.	.
acgn{2}cat	.	o	o	.	.
aacn{3}cat	.	o	o	.	.
aacn{8}aag	.	.	.	.	o
cctn{4}cac	.	.	o	.	.
aacn{0}gaa	.	o	o	.	.
actn{7}aag	.	.	o	.	.

Correctness statistics for all the genes having at least one annotated site in RegulonDB



Supplementary material

Discovered motifs in promoters of *glpD* orthologs from *Enterobacteriales*

Dyad	exp_freq	occ	exp_occ	occ_P	occ_E	sig	rank
atgn{1}tcg cgan{1}cat	0.00069	11	1.02	1.30E-08	2.40E-04	3.62	1
aacn{0}att aatn{0}gtt	0.00131	13	1.95	1.60E-07	3.00E-03	2.52	2
cgan{0}aca tgtn{0}tcg	0.00056	9	0.84	2.90E-07	5.20E-03	2.28	3
agtn{10}tcg cgan{10}act	0.00026	6	0.35	2.20E-06	3.90E-02	1.41	4
acan{13}acg cgtn{13}tgt	0.00044	7	0.6	3.50E-06	6.40E-02	1.19	5
aacn{2}aca tgtn{2}gtt	0.00082	9	1.21	5.60E-06	1.00E-01	0.99	6
aatn{2}tcg cgan{2}att	0.00089	9	1.31	1.00E-05	1.90E-01	0.73	7
atgn{0}ttc gaan{0}cat	0.00094	9	1.4	1.70E-05	3.10E-01	0.52	8
acgn{2}cat atgn{2}cgt	0.00054	7	0.79	2.10E-05	3.80E-01	0.42	9
aacn{3}cat atgn{3}gtt	0.00101	9	1.47	2.60E-05	4.60E-01	0.33	10
aacn{8}aag cttn{8}gtt	0.00080	8	1.11	2.60E-05	4.70E-01	0.33	11
cctn{4}cac gtgn{4}agg	0.00023	5	0.34	2.90E-05	5.30E-01	0.28	12
aacn{0}gaa ttcn{0}gtt	0.00106	9	1.58	4.40E-05	7.90E-01	0.1	13
actn{7}aag cttn{7}agt	0.00027	5	0.37	4.90E-05	8.90E-01	0.05	14

;assembly # 1 aaangtt.... .aatgtt.... .aatnntcg.. ..atgntcg.. ..atgttc... ..atgnnnngtt ..atgnncgt. ...tgttcg.. ...tgtnnngtt ...tgtncgt.gttngttgttcgt.ttcgtt aaatgttcgtt	seed: atgntcg	13 words
	aacnttt	0.82
	aacatt.	3.38
	..cgannatt.	2.1
	..cgancat..	7.55
	...gaacat..	3.29
	aacnnncat..	1.63
	.acgnncat..	1.46
	..cgaaca...	4.58
	aacnnaca...	2.4
	.acgnaca...	1.32
	aacnaac....	1.05
	.acgaac....	0.47
	aacgaa.....	1.53
	aacgaacattt	7.55
;assembly # 2 aaangtt..... .aatgtt..... .aatnntcg..... ..atgntcg..... ..atgttc..... ..atgnnnnnntcg. ..atgnnnngtt..... ..atgnncgt..... ..atgnnnnnnnncga ...tgttcg..... ...tgtnnngtt..... ...tgtncgt..... ...tgtnnnnntcg.gttngtt.....gttcgt.....gttnnnntcg.ttcgtt.....tcgnntcg.tcgnnnncga aaatgttcgttntcga	seed: atgnnnnnnnntcg	19 words
	aacnttt	0.82
	aacatt.	3.38
	cgannatt.	2.1
	cgancat..	7.55
	gaacat..	3.29
	.cgannnnnnncat..	1.82
	aacnnncat..	1.63
	acgnncat..	1.46
	tcgnnnnnnnncat..	0.93
	cgaaca...	4.58
	aacnnaca...	2.4
	acgnaca...	1.32
	.cgannnnnnnaca...	0.74
	aacnaac....	1.05
	acgaac....	0.47
	.cgannnnnaac....	0.35
	aacgaa.....	1.53
	.cgannncga.....	0.6
	tcgnnnncga.....	0.06
	tcganaacgaacattt	7.55
;assembly # 3 agtnnnnnnnnnntcg .gttnnnnnnnnnntcg ..ttcnnnnnnnnntcg agttcnnnnnnnnntcg	seed: agtnnnnnnnnnntcg	3 words
	cgannnnnnnnnnact	1.52
	cgannnnnnnnnaac.	0.9
	cgannnnnnnnngaa..	0.78
	cgannnnnnnnngaact	1.52
;assembly # 4 acannnnnnnnnnnnnacg. acannnnnnnnnnnnncga acannnnnnnnnnnnnaac.. acannnnnnnnnnnnnacga	seed: acannnnnnnnnnnnnacg	3 words
	.cgtnnnnnnnnnnnnntgt	1.26
	tcgnnnnnnnnnnnnnntgt	0.48
	..gttnnnnnnnnnnnntgt	0.01
	tcgttnnnnnnnnnnnntgt	1.26